WORKBOOK 04 / 08 · MULTIMEDIA 12

Walk to the edge with weight

Contact, down, passing, up, cycles and planted feet

24-FRAME WALK · EIGHT DESIGNED POSES



1 · contact | 4 · down | 7 · passing | 10 · up | 13 · contact | 16 · down | 19 · passing | 22 · up

Loop frames 1–24. Frame 25 matches frame 1 only for construction.

Eight designed poses for a 24-frame cycle. The optional construction endpoint at 25 is not an extra loop hold.

Build: Create a readable walk cycle and stage it without obvious foot sliding.

Keep: Walk_24 Graphic + staged approach + one diagnosed foot-slip repair

Shared story: A Caveman wants a third counting stone. A stream stands in his way. He studies it, commits to a jump, retrieves the stone and returns to count his collection with enormous pride.

Document: ActionScript 3.0 · 1280 × 720 · 24 fps · three main Scenes · Graphic symbols for scrubbable character actions

Two provisional 70-minute sessions. Use the HTML guide for read-aloud, switchable supports, recovery files and narrated explanation. This print version includes the essential support prompts.

Continue the same film

Create the project in Workbook 1. For every later workbook, open the selected previous production FLA and Save As a new numbered checkpoint, preserving all three Scenes and the complete Library. Keep repair/director copies separate; return to the chosen production version.

Understand before you animate

A walk alternates support. The body moves over a planted foot while the other leg swings forward.

The eight design poses are contact, down, passing and up, then the opposite leg repeats them. In a 24-frame cycle use poses at 1, 4, 7, 10, 13, 16, 19 and 22. A matching pose at 25 can help construct the seam, but the repeating export/loop contains frames 1–24 so the endpoint is not held twice.

Today’s checklist

1. Predict the result. 2. Build four stages. 3. Check in playback. 4. Refine, diagnose and direct. 5. Save and reopen.

Independent director condition

After retrieving the stone, the Caveman takes a careful departure walk along the far bank before the return-to-cave cut. Adjust stride, balance and hand contact; do not put a third-stone carry before he has acquired it.

What you will defend

Point to the supporting foot and show how its world position stays consistent during stance.

Support choices

Use the step summaries, glossary, numbered checklist, different-object examples, explanation frames and optional stretch. Open the companion HTML for browser read-aloud and the narrated six-slide explanation. Supports change access to the task; the success criteria remain the same.

Your planning notes

01 · Mark support before moving parts

SELECT: A new Walk_24 Graphic; guide layer above the character.

TOOLS / ROUTE: Timeline keys; guide line; Onion Skin.

1. Draw a ground guide and mark hip height. Add a contact reference at frame 1 and opposite contact at frame 13.

2. Name the supporting leg at each key. Put the feet on the same ground plane and make the forward/back separation readable.

3. Use near/far identity consistently. The near leg does not become the far layer when it travels behind the other leg.

WHY · Support and identity must stay coherent through the cycle.

CHECK / SUMMARY · Both contact poses have clear foot placement and opposite leg arrangements.

FIX · If legs switch visual identity, restore their original layer names and use depth ordering consciously.

GLOSSARY · contact pose: A key moment when a foot meets the ground.; stance: The phase in which a foot supports weight on the ground.

DIFFERENT-OBJECT EXAMPLE · A table does not change which leg it is when viewed from another angle; limb identity stays consistent too.

STRETCH · Record a short observation of your own walk without collecting anyone else's image.

02 · Build down, passing and up

SELECT: Walk_24 frames 4, 7 and 10; then 16, 19 and 22.

TOOLS / ROUTE: Insert Keyframe on all part layers; Free Transform.

1. At frame 4 lower the hips slightly and bend the supporting knee. At 7 bring the swing foot past the support leg with toe clearance. At 10 raise the hips slightly as the support heel lifts.

2. At frames 16, 19 and 22 repeat these jobs with opposite legs. Keep head movement smaller than hip movement.

3. Counter-swing the arms: the arm opposite the forward leg comes forward. Avoid identical rotations on both sides.

WHY · The rise and fall comes from support mechanics; it is not a random vertical bob.

CHECK / SUMMARY · The swing foot clears the ground and the body height changes are restrained and rhythmic.

FIX · If knees pop backward, inspect the joint bend direction and add a passing breakdown with a clear knee shape.

GLOSSARY · passing pose: The swing leg passes the supporting leg.; up pose: The higher body position as support approaches push-off.

DIFFERENT-OBJECT EXAMPLE · A two-legged toy looks weightless if its torso never responds to support changes.

STRETCH · Make the proud Caveman’s chest slightly lead while keeping the pelvis balanced.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

03 · Fill the cycle and inspect the seam

SELECT: The gaps between the eight design poses.

TOOLS / ROUTE: Insert Keyframe; Onion Skin; loop playback.

1. Add the required in-between poses or short classic spans between the designed keys. Check feet and knees on every frame where interpolation is used.

2. Create a matching frame 25 temporarily to compare with frame 1. Compare pose and movement direction across 22→24→1.

3. Keep a separate 25-frame construction copy. If a classic span ends at 25, bake that span to frame-by-frame through 24 before removing the construction endpoint, or manually key 23 and 24. The final loop contains frames 1–24 with no duplicate endpoint hold.

WHY · A cycle must join in both pose and rhythm; matching only the first and last picture is not enough.

CHECK / SUMMARY · No pause or snap occurs at the loop boundary.

FIX · If there is a hitch, remove a duplicate contact hold or repair spacing across the boundary. Do not hide the seam under a camera cut before diagnosing it.

GLOSSARY · cycle: A repeatable motion sequence.; loop seam: The transition from the final frame back to the first.

DIFFERENT-OBJECT EXAMPLE · A rotating wheel can have identical endpoint pictures but still pause if both are held in the loop.

STRETCH · Test two consecutive cycles with the ground guide visible.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

04 · Stage the approach in Crossing

SELECT: Main Scene Crossing; Character layer.

TOOLS / ROUTE: Library → drag Walk_24 instance; Graphic → Loop; main timeline keys.

1. Place a Walk_24 Graphic on the near bank and set Loop. Give the main Scene enough frames for three cycles: 72 frames at 24 fps.

2. Move the action container from a starting point to the near-bank stopping point using keys at 1 and 73 as construction endpoints. Right-click the intervening span and Create Classic Tween, then inspect the travel and Graphic playback. Bake through frame 72 before replacing the construction endpoint at 73 with Stop, so the final tween does not lose its endpoint.

3. Keep overall scale constant during the walk. Do not translate a flattened still and call it a walk cycle.

WHY · The inner action supplies articulation; the outer motion supplies travel across the scene.

CHECK / SUMMARY · Leg motion and overall travel work together, and the Caveman reaches the bank rather than crossing the gap during the walk.

FIX · If the inner action freezes, check Graphic Loop and parent timeline length. If it races, inspect the fps and duplicated endpoint.

GLOSSARY · nested animation: An action timeline placed inside another timeline.; screen translation: Movement of the action container across the Stage.

DIFFERENT-OBJECT EXAMPLE · A bicycle wheel rotates inside a bicycle that also travels across the screen.

STRETCH · Compare a slow cycle with short travel against a fast cycle with long travel.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

05 · Match stride and travel to reduce sliding

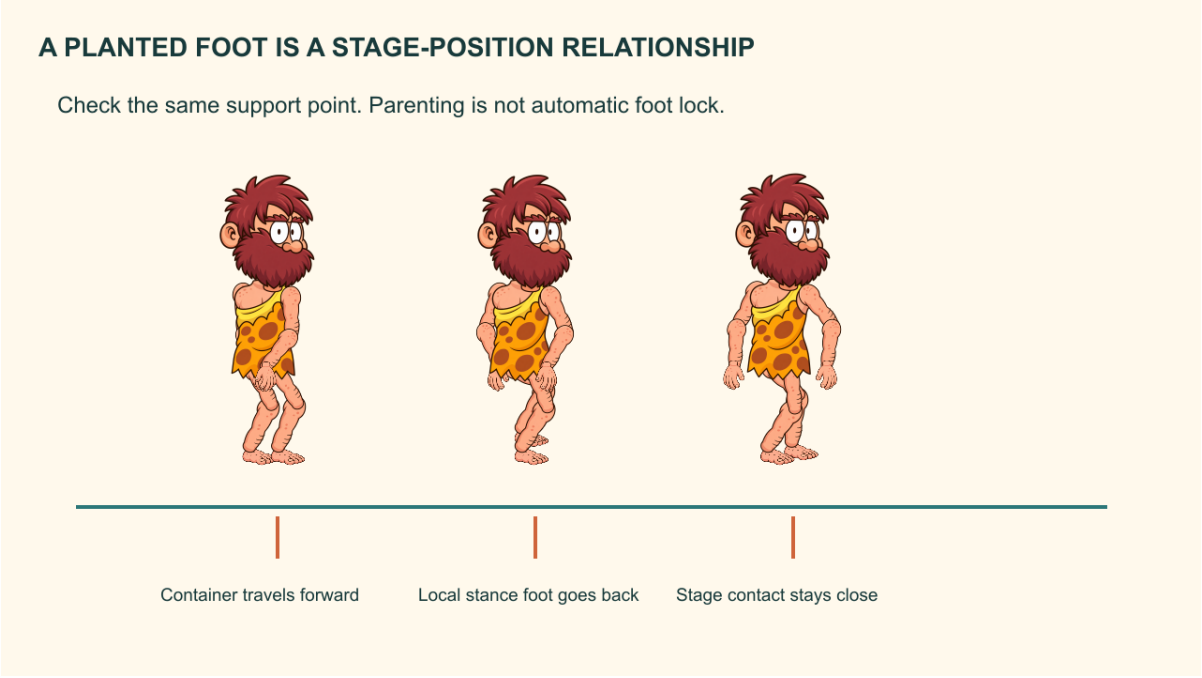
SELECT: A visible ankle/toe point during the stance half-cycle.

TOOLS / ROUTE: Onion Skin; frame-by-frame preview; guide marks.

1. Mark the supporting toe on the Stage at the start of stance. Advance several frames and compare its world position, including the outer container movement.

2. For the worked 24-frame cycle, at an unrotated 100% outer scale, a 12-frame stance with about 43 pixels of outer travel needs about 43 pixels of backward local foot travel to keep that foot planted. These are adjustable starting values, not universal anatomy. At 50% outer scale, about 86 local pixels produce 43 Stage pixels; judge the final Stage contact directly.

3. Adjust the local foot path, outer travel or stride duration together. Keep the foot at ground height through stance; lift it during swing.



Local foot movement and outer travel must agree during stance. Schematic comparison; assess actual Stage contact in playback.

WHY · A planted foot is a relationship between local leg motion and overall travel. Parenting alone does not lock it.

CHECK / SUMMARY · The support toe stays close to its guide mark until lift-off while the hips pass over it.

FIX · If it drifts forward, reduce outer speed or increase backward local foot motion. If it drifts backward, do the reverse. Recheck the other leg.

GLOSSARY · local position: A position within the action symbol.; world/Stage position: The final position after parent transforms are included.; foot slip: Unintended movement of a supposedly planted foot.

DIFFERENT-OBJECT EXAMPLE · A treadmill walker moves feet backward locally while staying in one place globally.

STRETCH · Measure slip in pixels over a stance and reduce it through one justified adjustment.

06 · Give the walk a motivated stop

SELECT: End of approach on the near bank.

TOOLS / ROUTE: Create a named 12-frame Graphic Stop from the neutral template; local keys and holds.

1. Create a Graphic named Stop with 12 local frames. Match local frame 1 to the final walk contact, shorten the final step, then plant the support foot before settling hips/chest over it. Use this named action at Crossing frames 73–84 in the worked assembly.

2. Let the arms finish their swing after the main body slows. Turn the head toward the far-bank stone.

3. Hold the assessment pose briefly before jump anticipation begins. The stop must not overlap an accidental looping walk frame.

WHY · Stopping needs a transfer of weight and a reason; an abrupt frozen cycle reads as an error.

CHECK / SUMMARY · The Caveman visibly judges the gap before preparing the jump.

FIX · If a foot pops, choose a compatible contact pose as the hand-off and repose the transition. Do not change the whole character's scale to reach the ground.

GLOSSARY · follow-through: A smaller continuation after the main action slows or stops.

DIFFERENT-OBJECT EXAMPLE · A runner's arms and clothing can finish moving after the feet stop.

STRETCH · Make a cautious stop and a confident stop with the same endpoint.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

07 · Repair one measured slip

SELECT: A duplicate approach with deliberately excessive outer travel.

TOOLS / ROUTE: Main timeline travel key; guide marks; before/after export.

1. Increase outer travel enough to create an obvious slide. Mark the support toe at three stance frames.

2. Restore plausible travel or revise the local stance path. Keep the cycle length unchanged for this comparison.

3. Retain three before/after toe positions and state which relationship you changed.

WHY · The repair shows whether you can diagnose the nested motion rather than just polish its appearance.

CHECK / SUMMARY · The repaired support point moves less while the body still reaches the intended stop.

FIX · If one foot improves but the other worsens, check the opposite half-cycle and consistent stride lengths.

GLOSSARY · constraint: A relationship that must be maintained, such as foot contact.

DIFFERENT-OBJECT EXAMPLE · A rolling wheel slips if its rotation and travel distance disagree.

STRETCH · Add a heel-to-toe roll while preserving the support-point logic.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

08 · Direct the carrying walk

SELECT: Your director action duplicate; stone attached by deliberate placement.

TOOLS / ROUTE: Pose keys; hand/prop layers; main scene travel.

1. After pickup, keep the stone held close while the Caveman walks carefully along the far bank. Reduce stride or hip bounce and revise balance rather than only changing playback speed.

2. Check hand-to-stone contact on every designed pose and stance-foot contact during travel.

3. Save and reopen. Explain one performance change and one contact repair, then use the carrying version only for the optional far-bank departure after pickup. Retain the empty-handed approach before the jump.

WHY · Carrying changes body mechanics and intention; it requires a new performance decision.

CHECK / SUMMARY · The prop stays held and the walk reads as careful under the same story goal.

FIX · If the prop floats, key it with the carrying hand or place it in the appropriate shared container and verify that relationship.

GLOSSARY · secondary action: A supporting action that enriches the main action without obscuring it.

DIFFERENT-OBJECT EXAMPLE · Independent task: use support, contact and silhouette criteria without a supplied carrying cycle.

STRETCH · Add a brief glance at the stone while preserving safe foot placement.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

Keep a small, useful evidence record

At frame ___, I chose ___ so the viewer could ___.
The problem was caused by ___. I changed ___ and checked ___.
Under the changed condition, I revised ___ because ___.

One repair

One independent decision and its trade-off

Delivery check

The selected file reopens. The result plays as intended. Joints/contact and audio were checked where relevant. The changed condition is solved. Required production files and a brief defence are retained.

Method references and attribution

A1 · Adobe Animate: symbols and instances · Adobe reference

A2 · Adobe Animate: timeline layers · Adobe reference

A3 · Adobe Animate: shape tweening · Adobe reference

A4 · Adobe Animate: parenting view · Adobe reference

A5 · Adobe Animate: sound · Adobe reference

A6 · Adobe Animate: camera · Adobe reference

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